

Interview Questions:

- Explain RAG project end-to-end including all components including system design and flow. Doc Loaders, Embedding models, Vector DB, Chunking Strategies, Retriever types, Retriever eval metrics, re-ranking procedure, Generator model metrics, MRR, STS score.
- How do you handle time-series data instead of text data for RAG, design how you store and retrieve.
- How do you improve accuracy of RAG model both retriever and generator.
- What evaluation frameworks are used for RAG? (RAGAS, DeepEval)
- What libraries have you used to extract tabular data and image data(Document AI) and how have you stored and retrieved them from vector DB in the context of RAG?
- Design RAG system with all services on a cloud where you upload new PDF file and it should automatically extract and store info in vectorDB, how will you do it and it should also scale automatically.
- Explain about your understanding of agents and why do we need them?
- Explain what is Runnable in langchain.
- How to integrate langsmith into langgraph.
- Working of multi-agents and project explanation of agents wrt to BES
- How will you route requests to right model and write code with proper syntax and real world example of it.
- If an agent encounters FAILED msg at end, how will agent know to retry and trigger workflow from start and send response, implement this.
- Imagine you have 100 GB file, with limited RAM how will you extract lines containing ERROR keyword in python?
- Dict comprehension, Dict sorting with None values, Lambda functions, Map, Filter, Generator and Decorators.
- Multithreading vs Multiprocessing. Imagine you have 2 processes, one is blocking and non-blocking process, what is the efficient way to execute them.
- About kubeflow, Airflow, DVC, MLOPs, LLMOPs.
- Implement state in agentic system with proper syntax for message storage using TypedDict.
- Key, Query, Value how it works mathematically and how attention works in general. How are these scores calculated.
- Extract images from pdf using langchain or any other libraries.
- Why do you use base64 format for images, uses
- Deploying agent into PROD, what kind of challenges we would face and how to overcome them
- Git Rebase vs Squash vs Pull
- Agentic AI to PROD how to build CICD and what all thinks to include in CI pipeline.
- What is HPA (horizontal autopod scaler)
- Overlapping merge intervals problem? Implement FLAMES logic in python?
- Multiple inheritance and Multi-level inheritance
- Single Agent vs Multi Agent vs Deep Agent

- Guardrails and how do you implement this in langgraph, would you place guardrails at every node or tool routing node or tool level or how exactly and why.
- Design a gateway to ensure sensitive/personal information is not sent to LLM
- Any ORM tool used in python, how to you design CICD to make updations to database using ORM.
- What does metrics does RAGAS framework outputs and ROUGE score, MRR explanation and formula.
- How does agents pass context in between multi agents?
- Usage of Guardrails in langgraph
- If you have 2 kinds of users premium and free user then design a gateway in agentic system such a way that premium user request should land at premium LLM model with larger output context window and free user should get any random open source model.
- Have you worked with injecting context into an invocation workflow in agents?
- How do you handle hallucinations if your context window is 100k length?
- Coding Questions: 1) Find indexes of pairs of numbers in list whose sum is equal to given target. 2) Find the 2nd largest number and 4th largest number in $O(n)$ complexity. 3) Find the duplicates in given list whose repetition is 2 times. 4) Find the missing numbers in given list whose range is 1 to n. 5) You have list of dict with retrieved chunks write code to find Recall at K and Precision at K.
- Langfuse and Langsmith, when and how to use.